DESCENDING PURITY ALONG AN INVERSE LIMIT OF AFFINE SCHEMES

THE STACKS PROJECT AUTHORS

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Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Openness of flatness and descent of finite presentation, modules, opens and flatness along a limit. More on Morphisms, label more-morphisms-theorem-openness-flatness (proof line 5397); Limits, labels limits-lemma-descend-finite-presentation (proof line 5401), limits-lemma-descend-modules-finite-presentation (proof line 5406), limits-lemma-descend-opens (proof line 5411) and limits-lemma-descend-module-flat-finite-presentation (proof line 5415). Makes the flat locus open so a quasi-compact open inside it can be found, and then descends the morphism, the module, the chosen open and the flatness of the module over a quasi-compact piece to some stage of the inverse system.
- An elementary etale neighbourhood on which the module has projective global sections, and purity of an affine locally projective module. Same chapter flat.tex, labels lemma-finite-presentation-flat-along-fibre (proof line 5422) and lemma-affine-locally-projective-pure (proof line 5436). Produces the affine etale neighbourhood S_i' -> S_i and the etale X_i' -> X_i whose global sections of the pulled back module are projective over the base, and gives that this pulled back module is universally pure over S_i'.
- The criterion for purity through associated primes and constructibility of the relative assassin condition. Same chapter flat.tex, labels lemma-criterion (proof lines 5452 and 5484); More on Morphisms, labels more-morphisms-lemma-relative-assassin-constructible (proof line 5461) and more-morphisms-lemma-base-change-assassin-in-U (proof line 5471); Limits, label limits-lemma-limit-contained-in-constructible (proof line 5482). Turns purity along the fibre into an inclusion of the relative associated primes

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inside the image of the etale morphism, makes the two sets of points where that inclusion holds locally constructible and compatible with base change, and allows the inclusion to be tested at a finite stage because the local ring of the limit is the filtered colimit of the local rings of the stages.

Source reference. Original lemma 05K6: sources/stacks-flat/pinned-original-source.tex, source lines 5148–5148. Navigation pointer for \ref{lemma-affine-locally-projective-pure} used inside the counted span (line 5436); the argument that consumes it is retained in the span.

Source reference. Original lemma 05L4: sources/stacks-flat/pinned-original-source.tex, source lines 5197–5197. Navigation pointer for \ref{lemma-criterion} used inside the counted span (line 5452); the argument that consumes it is retained in the span.

Source reference. Original lemma 05KW: sources/stacks-flat/pinned-original-source.tex, source lines 3342–3342. Navigation pointer for \ref{lemma-finite-presentation-flat-along-fibre} used inside the counted span (line 5422); the argument that consumes it is retained in the span.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Navigation pointer for \ref{limits-lemma-descend-finite-presentation} used inside the counted span (line 5401); the argument that consumes it is retained in the span.

Source reference. Original lemma 05LY: sources/stacks-chow/tags.txt, source lines 7252–7252. Navigation pointer for \ref{limits-lemma-descend-module-flat-finite-presentation} used inside the counted span (line 5415); the argument that consumes it is retained in the span.

Source reference. Original lemma 01ZR: sources/stacks-chow/tags.txt, source lines 2604–2604. Navigation pointer for \ref{limits-lemma-descend-modules-finite-presentation} used inside the counted span (line 5406); the argument that consumes it is retained in the span.

Source reference. Original lemma 01Z4: sources/stacks-chow/tags.txt, source lines 2579–2579. Navigation pointer for \ref{limits-lemma-descend-opens} used inside the counted span (line 5411); the argument that consumes it is retained in the span.

Source reference. Original lemma 05F4: sources/stacks-chow/tags.txt, source lines 7013–7013. Navigation pointer for \ref{limits-lemma-limit-contained-in-constructible} used inside the counted span (line 5482); the argument that consumes it is retained in the span.

Source reference. Original lemma 05KQ: sources/stacks-chow/tags.txt, source lines 7209–7209. Navigation pointer for \ref{more-morphisms-lemma-base-change-assassin-in-U} used inside the counted span (line 5471); the argument that consumes it is retained in the span.

Source reference. Original lemma 05KR: sources/stacks-chow/tags.txt, source lines 7210–7210. Navigation pointer for \ref{more-morphisms-lemma-relative-assassin-constructible} used inside the counted span (line 5461); the argument that consumes it is retained in the span.

Source reference. Original lemma 0399: sources/stacks-chow/tags.txt, source lines 4264–4264. Navigation pointer for \ref{more-morphisms-theorem-openness-flatness} used inside the counted span (line 5397); the argument that consumes it is retained in the span.

Lemma 0.1. *Let I be a directed set. Let $(S_i, g_{ii'})$ be an inverse system of affine schemes over I. Set $S = \lim_i S_i$ and $s \in S$. Denote $g_i : S \to S_i$ the projections and set $s_i = g_i(s)$. Suppose that $f : X \to S$ is a morphism of finite presentation, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$-module of finite presentation which is pure along X_s and flat over S at all points of X_s. Then there exists an $i \in I$, a morphism of finite presentation $X_i \to S_i$, a quasi-coherent $\mathcal{O}_{X_i}$-module $\mathcal{F}_i$ of finite presentation which is pure along $(X_i)_{s_i}$ and flat over S_i at all points of $(X_i)_{s_i}$ such that $X \cong X_i \times_{S_i} S$ and such that the pullback of $\mathcal{F}_i$ to X is isomorphic to $\mathcal{F}$.*

Proof. Let $U \subset X$ be the set of points where $\mathcal{F}$ is flat over S. By More on Morphisms, Theorem 0399 this is an open subscheme of X. By assumption $X_s \subset U$. As X_s is quasi-compact, we can find a quasi-compact open $U' \subset U$ with $X_s \subset U'$. By Limits, Lemma 01ZM we can find an $i \in I$ and a morphism of finite presentation $f_i : X_i \to S_i$ whose base change to S is isomorphic to f_i. Fix such a choice and set $X_{i'} = X_i \times_{S_i} S_{i'}$. Then $X = \lim_{i'} X_{i'}$ with affine transition morphisms. By Limits, Lemma 01ZR we can, after possible increasing i assume there exists a quasi-coherent $\mathcal{O}_{X_i}$-module $\mathcal{F}_i$ of finite presentation whose base change to S is isomorphic to $\mathcal{F}$. By Limits, Lemma 01Z4 after possibly increasing i we may assume there exists an open $U_i' \subset X_i$ whose inverse image in X is U'. Note that in particular $(X_i)_{s_i} \subset U_i'$. By Limits, Lemma 05LY (after increasing i once more) we may assume that $\mathcal{F}_i$ is flat on U_i'. In particular we see that $\mathcal{F}_i$ is flat along $(X_i)_{s_i}$.

Next, we use Lemma 05KW to choose an elementary étale neighbourhood $(S_i', s_i') \to (S_i, s_i)$ and a commutative diagram of schemes

$$\begin{array}{ccc} X_i & \xleftarrow{g_i} & X_i' \\ \downarrow & & \downarrow \\ S_i & \longleftarrow & S_i' \end{array}$$

such that g_i is étale, $(X_i)_{s_i} \subset g_i(X_i')$, the schemes X_i', S_i' are affine, and such that $\Gamma(X_i', g_i^*\mathcal{F}_i)$ is a projective $\Gamma(S_i', \mathcal{O}_{S_i'})$-module. Note that $g_i^*\mathcal{F}_i$ is universally pure over S_i', see Lemma 05K6. We may base change the diagram above to a diagram with morphisms $(S_{i'}', s_{i'}') \to (S_{i'}, s_{i'})$ and $g_{i'} : X_{i'}' \to X_{i'}$ over $S_{i'}$ for any $i' \geq i$ and we may base change the diagram to a diagram with morphisms $(S', s') \to (S, s)$ and $g : X' \to X$ over S.

At this point we can use our criterion for purity. Set $W_i' \subset X_i \times_{S_i} S_i'$ equal to the image of the étale morphism $X_i' \to X_i \times_{S_i} S_i'$. For every $i' \geq i$ we have similarly the image $W_{i'}' \subset X_{i'} \times_{S_{i'}} S_{i'}'$ and we have the image $W' \subset X \times_S S'$. Taking images commutes with base change, hence $W_{i'}' = W_i' \times_{S_i'} S_{i'}'$ and $W' = W_i \times_{S_i'} S'$. Because $\mathcal{F}$ is pure along X_s the Lemma 05L4 implies that

$$f^{-1}(\mathrm{Spec}(\mathcal{O}_{S',s'})) \cap \mathrm{Ass}_{X \times_S S'/S'}(\mathcal{F} \times_S S') \subset W' \tag{0.1.1}$$

By More on Morphisms, Lemma 05KR we see that

$$E = \{t \in S' \mid \mathrm{Ass}_{X_t}(\mathcal{F}_t) \subset W'\} \quad \text{and} \quad E_{i'} = \{t \in S_{i'}' \mid \mathrm{Ass}_{X_t}(\mathcal{F}_{i',t}) \subset W_{i'}'\}$$

are locally constructible subsets of S' and $S_{i'}'$. By More on Morphisms, Lemma 05KQ we see that $E_{i'}$ is the inverse image of E_i under the morphism $S_{i'}' \to S_i'$ and that E is the inverse image of E_i under the morphism $S' \to S_i'$. Thus Equation (0.1.1) is equivalent to the assertion that $\mathrm{Spec}(\mathcal{O}_{S',s'})$ maps into E_i. As $\mathcal{O}_{S',s'} = \mathrm{colim}_{i' \geq i} \mathcal{O}_{S_{i'}', s_{i'}'}$ we see that $\mathrm{Spec}(\mathcal{O}_{S_{i'}', s_{i'}'})$ maps into E_i for some $i' \geq i$, see Limits, Lemma 05F4. Then, applying Lemma 05L4 to the situation over $S_{i'}$, we conclude that $\mathcal{F}_{i'}$ is pure along $(X_{i'})_{s_{i'}}$. □